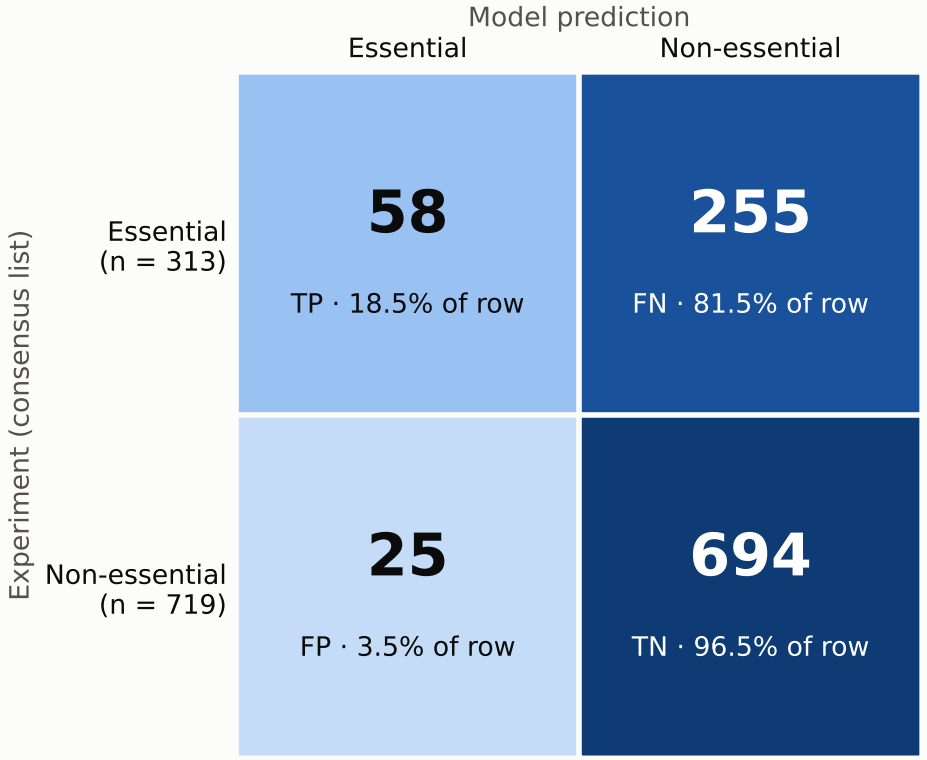
Single-gene essentiality vs consensus essential list (KO/WT < 15 %)

Task A medium + PO1f, glucose uptake 2.414 mmol/gDW/h

iYali21 (iyli21.xml, 6974b758)



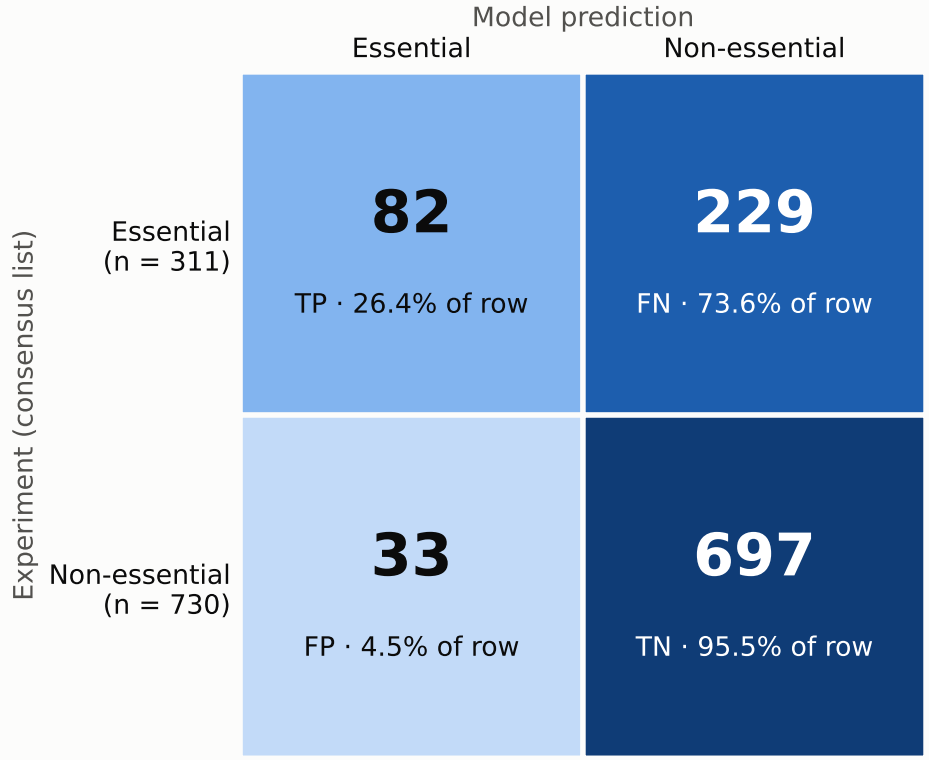
Recall 18.5% · Precision 69.9% · Specificity 96.5% · MCC 0.254

WT 0.2982 h⁻¹ · 1,058 genes screened · 16 mtDNA excluded · 10 unresolved

If no-solution KOs = lethal: TP 65 · FN 255 · FP 28 · TN 694 · MCC 0.266

Listed positives in model 320 / 1,612 (1,292 out of scope, not FN)

iYali26 newest candidate (a56f3649)



Recall 26.4% · Precision 71.3% · Specificity 95.5% · MCC 0.319

WT 0.3124 h⁻¹ · 1,073 genes screened · 16 mtDNA excluded · 16 unresolved

If no-solution KOs = lethal: TP 93 · FN 229 · FP 38 · TN 697 · MCC 0.331

Listed positives in model 322 / 1,612 (1,290 out of scope, not FN)

Static FBA, Task A medium (po1f_csm_leu_w29uptake_20261012.csv, glucose uptake 2.414) + PO1f strain profile, objective biomass_C; essential = unrounded KO/WT strictly below 0.15.
Matrix cells follow the benchmark rule: a knockout with no LP solution is unresolved and left out. Cell shade = share of the experimental row.
Negatives are model genes not on the consensus list (a user-defined class, not experimental non-essential calls). The list was used during model development,
so this is a development reference, not independent validation. Screens run 2026-10-14.